

Detailed ProteoMIX data processing workflow (DIA-charge reduction + SLOMO-nMS)



nTD prioritization based on function

Data Processing (Script 1):

- Identical scans are averaged across duty cycles (fwd sliding window averaging, e.g. N=5 cycles).
- Unidec deconvolves averaged spectra from the time-course.
- Peak picking:
 - Proteoform peaks are selected for downstream analysis following filtering by:
 - D-score (e.g. ≥ 0.4).
 - Intensity (≥ 0.03 , normalized to 1 for respective scan).
 - Number of times it appears in different isolation windows (e.g. ≥ 2).

Script 1 relevant output:

- averaged raw scans ("`<fname>/averaged_scan<number>_cyc<number>.txt`")
- deconvolved mass spectra ("`<fname>/av_mass_scan<number>_cyc<number>.txt`")
 - filtered proteoform mass list ("`<fname>/<fname>_ppeaks.csv`")



Time-Series Analysis (Script 2):

- Input data: -- deconvolved mass spectra (or averaged scans) from Script 1 output.
- Peak picking by intensity from defined cycles or manual peak list (e.g. from Script 1 output).
- Time-series intensity extraction and data normalization.
- Time-courses from each extract are clustered. A correlation analysis is also performed.
- Clustering results and overlayed spectra for each proteoform are output to a .pdf.

Script 2 relevant output:

- PDF of results. Candidate ligand binding proteoforms:
 - sharp loss or gain in intensity during time-course.
 - strong anticorrelation with another proteoform (mass difference = candidate ligand).
- m/z's of candidate functional proteoform masses are in proteoform mass list from script 1 output.



nTD-verify functional proteoforms